

Smith et al. (2020)

Habitat Quality Assessment for Chinook Salmon

Interior British Columbia Streams

1. Spawning Substrate

Field surveys at 47 index sites found that reaches where embeddedness exceeded 25% had egg-to-fry survival rates 62% lower than reaches where embeddedness was less than 25%.

2. Riparian Shade and Stream Temperature

At sites where fish were harvested within 30 m of the bankfull channel, daily maximum temperatures increased by an average of 4.2 degrees C compared to sites where fish were harvested further from the channel.

3. Road Crossings and Fish Passage

Barriers to adult chinook migration. Culverts with outlet drops exceeding 15 cm or insufficient water depth during low flow accounted for 75% of the barriers.

4. Large Woody Debris

Correlated with large woody debris loading. Sites with more than 25 pieces of wood per 100 m of channel had 2.8 times higher juvenile chinook densities than sites with fewer than 25 pieces of wood per 100 m.